

Numerics II

Freie Universität Berlin

Winter semester 2025/2026

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Exercise Sheet 4

Deadline: Friday, November 14th, 2025 at 10 am

Exercise 1 (Stability function at infinity; 3 points)

Consider a Runge-Kutta method defined by the triple $(\mathbf{A}, \mathbf{b}, \mathbf{c})$, given in the Butcher tableau

$$\begin{array}{c|c} \mathbf{c} & \mathbf{A} \\ \hline & \mathbf{b}^\top \end{array},$$

with s stages and where it is assumed that $\mathbf{A}$ is non-singular.

Prove the following statement:

It holds $R(\infty) = 0$ if one of the following conditions is satisfied:

- (i) $a_{si} = b_i$ for $i = 1, \dots, s$, or
- (ii) $a_{i1} = b_1$ for $i = 1, \dots, s$, $b_1 \neq 0$.

Hint: Derive first a formula for $R(\infty)$.

Exercise 2 (Stage 2 Runge-Kutta schemes of consistency order 2; 4 points)

For simplicity let $n = 1$.

Runge-Kutta schemes of stage $s = 2$ have the incremental function

$$\Phi(t, Y, h) = b_1 f(x, Y) + b_2 f(t + c_2 h, Y + h a_{21} f(t, Y)) \in \mathbb{R}.$$

Show that Taylor expansion around $h = 0$ and requiring consistency order $q = s = 2$ yields the condition for $q = 2$ from Theorem 1.42.

Exercise 3 (Erwin Fehlberg's RK4(3) (cf. Example 1.55); 4 points)

Based on classical Runge-Kutta method of order $\tilde{q} = 4$ we consider the embedded tableau

$$\begin{array}{c|ccccc} 0 & 0 & \dots & \dots & \dots & 0 \\ \frac{1}{2} & \frac{1}{2} & & & & \vdots \\ \frac{1}{2} & 0 & \frac{1}{2} & & & \vdots \\ 1 & 0 & 0 & 1 & & \vdots \\ c_5 & a_{51} & a_{52} & a_{53} & a_{54} & 0 \\ \hline & b_1 & b_2 & b_3 & b_4 & b_5 \\ & \frac{1}{6} & \frac{1}{3} & \frac{1}{3} & \frac{1}{6} & 0 \end{array}.$$

Use the Fehlberg-Trick and Theorem 1.42 to determine the parameters that will lead to an embedded method of order $q = 3$.

Programming Exercise (The van der Pol system; $2 + 2 + 1 = 5$ points)

Consider the van der Pol system of differential equations given by

$$\begin{aligned} y_1'(t) &= y_2(t), \\ y_2'(t) &= 10(1 - y_1(t)^2)y_2(t) - y_1(t), \quad t \in (0, 20], \end{aligned}$$

with initial conditions

$$y_1(0) = 0 \quad \text{and} \quad y_2(0) = 1.$$

- a) Use your Python code `runge_kutta_ex(f, y0, I, h, weights, A)` from the programming exercise on sheet 3 together with the Butcher tableau of the classical fourth-order Runge-Kutta method and adjust it to solve the system above for $I = [0, 20]$ and step size $h \in \{0.5, 0.25, 0.125, 0.0625\}$. Plot $y_1(t)$ and $y_2(t)$ versus the time t .
- b) Since the van der Pol system has no closed-form solution, compute a reference solution numerically y_{ref} using the same Runge-Kutta scheme but with a very small step size $h_{\text{ref}} = 2^{-14}$. Halve h_{ref} once more to compute $\tilde{y}_{\text{ref}}$ and check that

$$\max_k \|y_{\text{ref}}(t_k) - \tilde{y}_{\text{ref}}(t_k)\|_{\infty} < 10^{-9}.$$

Use y_{ref} to compute

$$\max_k \|Y_k - y_{\text{ref}}(t_k)\|_{\infty}.$$

- c) Describe your observations.

Rules for submission & passing criterion:

Exercises must be completed in **groups of three** and **submitted electronically via the Whiteboard system** under “**Assignments**.” Late submissions will not be corrected. Please **state the names of all group members** on the submission. The programming language of this course is **Python**. The exercises will be discussed in the tutorial the following week. To pass the tutorial component of the module, you must achieve **at least 50% of the homework points**.